UX CASE STUDY · PRODUCT MANAGEMENT

Aura: from crowded to calm

Redesigning a spaced-repetition flashcard app around one clear next action

Solo PM · Hands-on audit, proto-personas, journey mapping, prioritization, shipped fixes · November 2026



THE SHORT VERSION

Aura had every feature on every screen, except a clear way to study

PROBLEM

Crowded, and hard to trust

Voice and AI prompts sat on every surface. The core loop, reviewing what's due, was buried, and the numbers on screen contradicted each other.

WHAT I DID

Audit, model, prioritize

Tested every non-AI screen on an iPhone simulator, built three proto-personas, mapped their journeys and ranked fixes by impact and effort.

OUTCOME

Ship now, redesign next

Quick wins shipped in one PR: honest counts, an answer-reveal step, grading in thumb reach. A v2 "Today" flow is ready to test.

CONTEXT

A flashcard app with three products inside it

Spaced repetition. Decks of cards, graded Again / Hard / Good / Easy, rescheduled with an SM-2 style interval.

AI card generation. Paste notes and get draft cards with readings, mnemonics and example sentences.

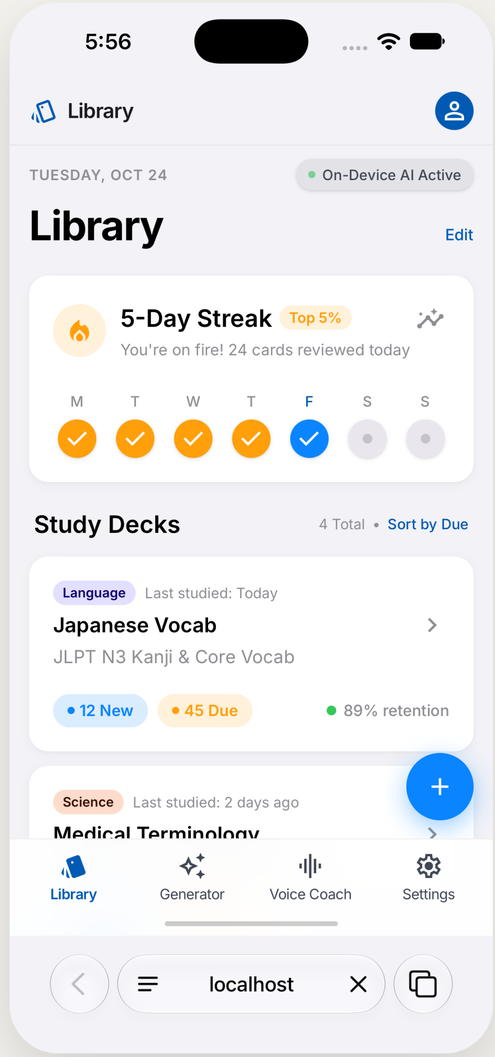
Voice coach. Say the answer out loud and get spoken feedback on recall and pronunciation.

My role

Solo PM with hands-on prototyping: audit, research synthesis, specs and the fix PR.

Scope

The study loop, decks and settings. AI generation and voice quality were out of scope.



APPROACH

Evidence first, then people, then priorities

1

Hands-on audit

Every non-AI screen on an iPhone simulator, plus a read of the code behind each state.

2

Proto-personas

Three learners the issues hurt most, each with a job to be done.

3

Journey maps

Five stages per persona, to find where each one stalls or quits.

4

Prioritize

Impact against effort. Quick wins shipped straight away.

5

Propose v2

Bigger flow changes, framed as hypotheses to test with real users.

⚠ **Honest caveat:** these personas are hypotheses from the audit, not interviews. The last slide covers validation.

AUDIT EVIDENCE

What a first session shows

1 Library

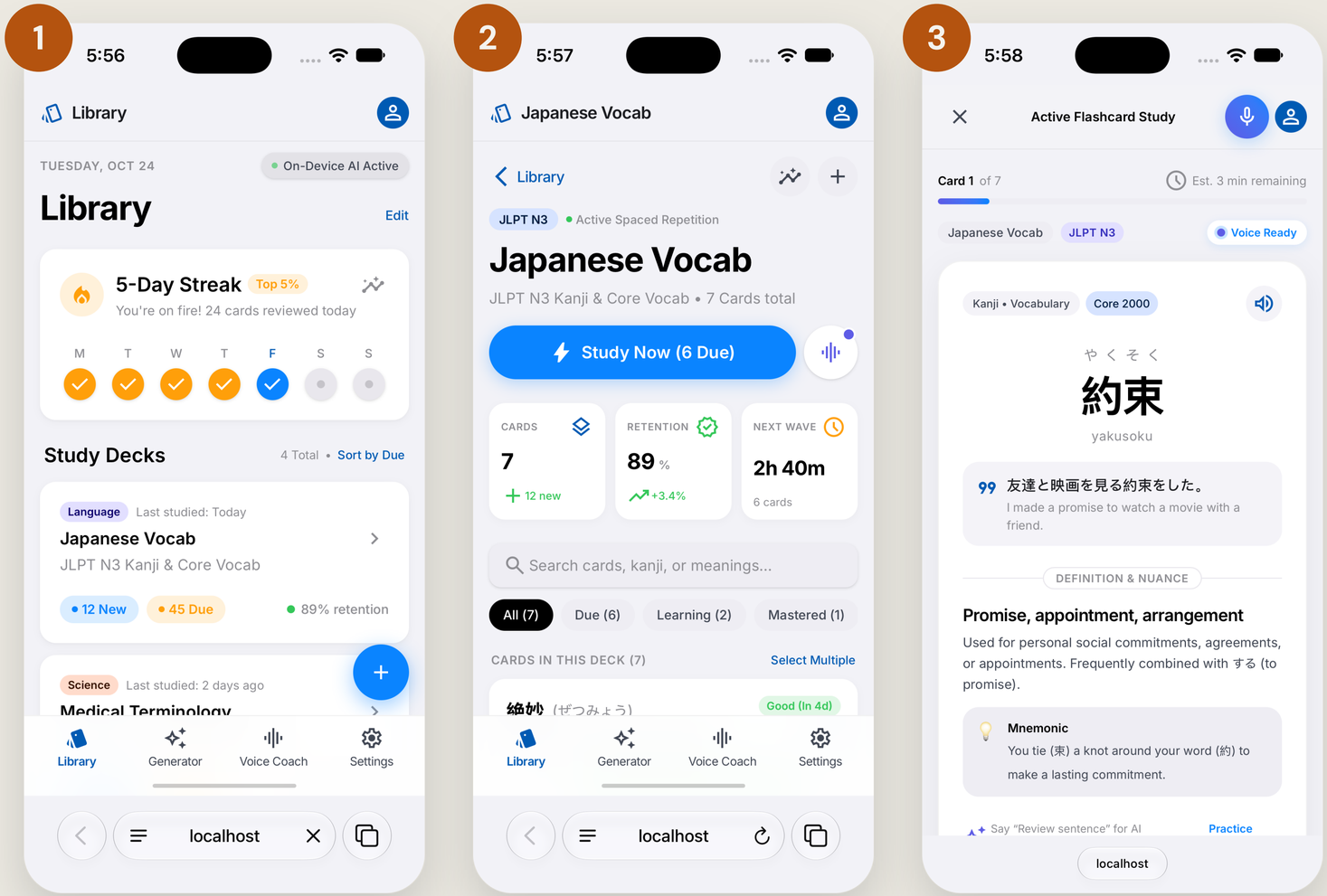
Japanese Vocab shows 12 new and 45 due. The deck holds 7 cards.

2 Deck detail

Same deck: 7 cards, 6 due. Two screens, two truths.

3 Study session

The answer sits under the question, so there's nothing left to recall. Grades are below the fold.



WHAT I FOUND

Four themes behind the issues

TRUST

The numbers disagree

Deck counts were stored apart from the cards, so screens drifted. “Due tomorrow” counted as due.

CORE LOOP

Recall without recalling

No reveal step, so every grade was a guess. On iPhone, the grade buttons sat below the fold.

NOISE

Voice and AI everywhere

4 voice or AI prompts on the home screen and 5 on the study screen, all competing with studying.

POLISH

Small frictions add up

A tiny Edit link, a + button over delete, Japanese-only fields on a medical deck.

M

Maya

The exam crammer

Proto-persona

21, third-year university student

Goal: pass JLPT N3 in 6 weeks

When: 10-minute bursts on the bus

How: phone in one hand, standing

JOB TO BE DONE

When I have ten minutes on the bus, I want to clear exactly what's due, so I walk into the exam having seen every card enough times.

Goals

- Clear every due card, every day
- Know if the pace will reach the exam
- Study with one thumb

Frustrations today

- Grades below the fold: two hands per card
- Answer visible before recall
- Tapping a card restarted at card 1

Needs from Aura

- Show Answer, then grades in thumb reach
- Swipe to grade
- Exam countdown with a daily target

Success: finishes the due queue in one bus ride, one-handed.

D

Daniel

The busy professional

Proto-persona

34, product manager

Goal: basic Japanese before a move to Tokyo in 4 months

When: 5 minutes over morning coffee

How: opens the app, wants to be studying at once

JOB TO BE DONE

When I have five minutes before work, I want to start reviewing in one tap, so a daily habit survives a full calendar.

Goals

- Keep a daily streak going
- One “am I on track” number
- Zero setup per session

Frustrations today

- Counts that change between screens
- 9 voice and AI prompts before the first card
- Streak copy that ignores today

Needs from Aura

- One due number, the same everywhere
- A single Start button on open
- Voice and AI only when asked for

Success: from opening the app to the first card in under 5 seconds.

P

Priya

The lapsed learner

Proto-persona

28, nurse on rotating shifts

Goal: enough Japanese for a trip next spring

When: irregular; just back after 3 weeks of night shifts

How: tired, low patience, easily put off

JOB TO BE DONE

When I come back after weeks away, I want a small, doable session, so I restart the habit instead of giving up on it.

Goals

- Get back in without guilt
- A session that fits one break
- Not lose earlier progress

Frustrations today

- The whole backlog at once, no cap
- “0 day streak” as the first message
- No idea how long the session is

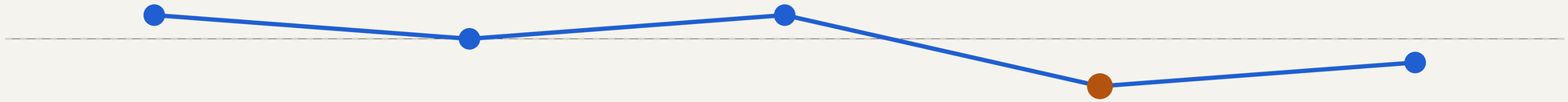
Needs from Aura

- A catch-up mode with a daily cap
- A friendly welcome back
- Session length shown up front

Success: completes a second session within a week of coming back.

JOURNEY MAP · MAYA, THE EXAM CRAMMER

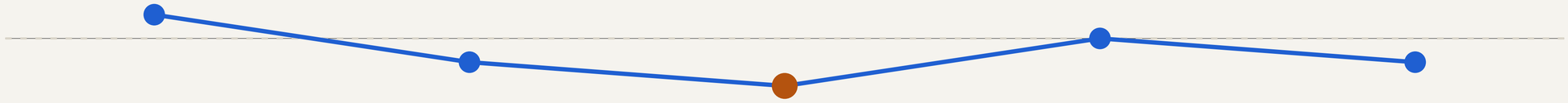
Ten minutes on the bus, lost at the review step

STAGE	Trigger	Orient	Start	Review	Finish
DOING	Boards the bus, opens Aura with one hand on the rail	Reads 45 due on a deck that holds 7 cards	Taps a hard card; the session starts at card 1	Answer already shown; scrolls down to reach grades	Stop arrives with no sense of what's left
FEELING					
	Motivated	Confused	Hopeful	Frustrated	Unsure
PAIN	Voice and AI banners above the decks	Can't plan 10 minutes around a wrong number	Repeats cards already seen today	No recall test; needs two hands to grade	No progress or end state
OPPORTUNITY	Open on Today, studying first	Counts derived from the cards	Start at the tapped card, due only	Show Answer, pinned grades, swipe	"All caught up" finish screen

JOURNEY MAP · DANIEL, THE BUSY PROFESSIONAL

Five minutes, mostly spent finding the start button

STAGE	Trigger	Orient	Start	Review	Finish
DOING	Coffee poured; opens Aura before work	Scans pills, banners and four tabs	Hunts for where to begin studying	Reviews a few cards; it works	Leaves unsure if today counted
FEELING	Ready	Overloaded	Impatient	Relieved	Unsure
PAIN	Nothing on the first screen says “start”	9 voice and AI prompts before a card	10 tap targets, none starts a review	Counts change between screens	Streak said 0 days, ignoring today
OPPORTUNITY	One-tap Start review on Today	Voice and AI moved out of the way	Due count and time estimate up front	One due number everywhere	Streak counts today’s session



JOURNEY MAP · PRIYA, THE LAPSED LEARNER

Back after three weeks, met with the whole backlog

STAGE	Trigger	Orient	Start	Review	Finish
DOING	Sees a trip reminder; reopens Aura	Sees the full overdue pile and a broken streak	Starts a session with no end in sight	Grades many forgotten cards as Again	Quits partway, unsure when to return
FEELING	Guilty	Overwhelmed	Reluctant	Discouraged	Defeated
PAIN	No welcome back, just numbers	Backlog shown all at once	No session size or time estimate	Failures pile up; no sense of progress	Nothing plans the next session
OPPORTUNITY	Welcome-back message after 7+ days	Catch-up mode: 20 today, rest over 5 days	Show “about 4 minutes” before starting	Progress bar; celebrate cards recovered	Remind about tomorrow’s small batch



USER STORIES

Six stories, each with a test for “done”

Maya · exam crammer

As a student studying one-handed, I want the grade buttons pinned in thumb reach, so I can grade without a second hand.

Done when: grades stay visible on a 6.3" screen without scrolling; swipe left/right grades Again/Good.

Shipped · Must

As a learner, I want the answer hidden until I ask for it, so each grade reflects real recall.

Done when: only the prompt shows first; Show Answer reveals meaning, notes and translation.

Shipped · Must

Daniel · busy professional

As a busy learner, I want one due number that matches everywhere, so I can trust how much is left today.

Done when: Library, deck detail and session size all show the same count, derived from the cards.

Shipped · Must

As a busy learner, I want to start reviewing in one tap from launch, so five minutes is enough for a session.

Done when: app opens on Today; Start review is the primary action; voice and AI live behind it.

v2 · Should

Priya · lapsed learner

As a returning learner, I want my backlog capped and spread out, so coming back feels doable.

Done when: after 7+ days away a session caps at 20 cards; the rest spreads over at most 5 days.

v2 · Must

As a returning learner, I want a welcome back instead of a broken streak, so I don’t feel punished for leaving.

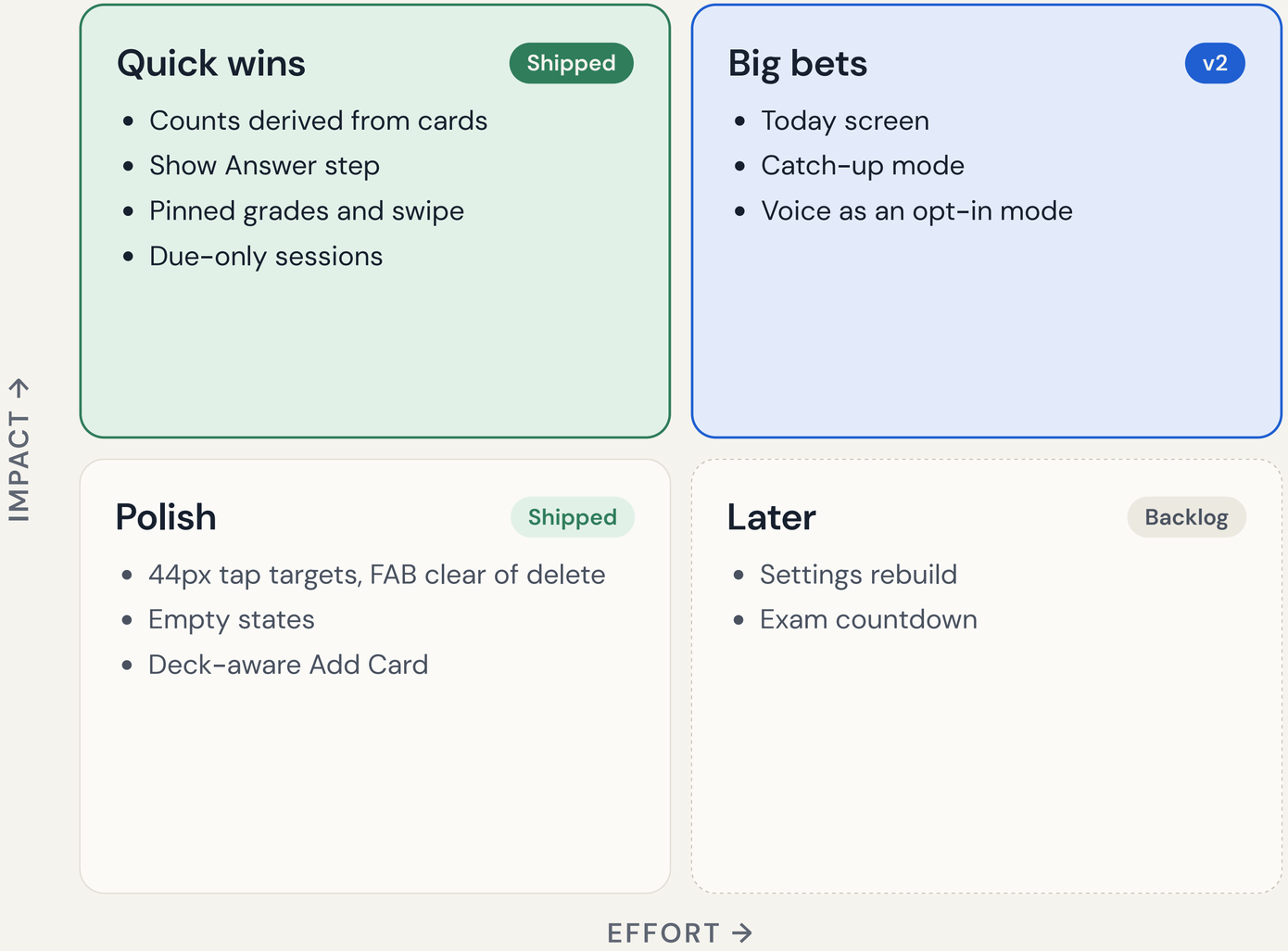
Done when: after 7+ days away, Today greets the learner and offers catch-up before any streak copy.

v2 · Could

PRIORITIZATION

Four principles, then impact vs effort

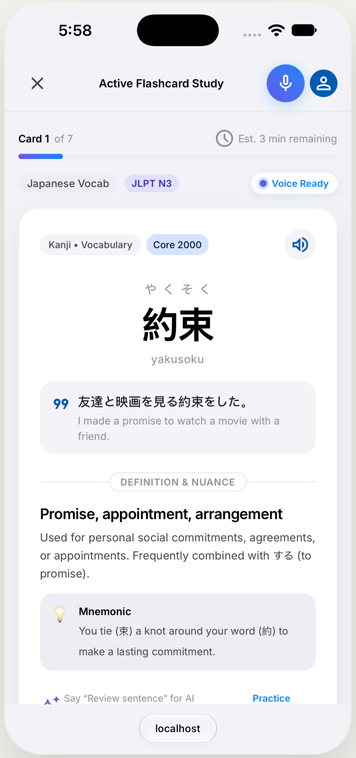
1. **One clear next action** on every screen
2. **Numbers you can trust**, computed from one source
3. **Calm by default**, power on demand
4. **Thumb-first**: key actions in the bottom third



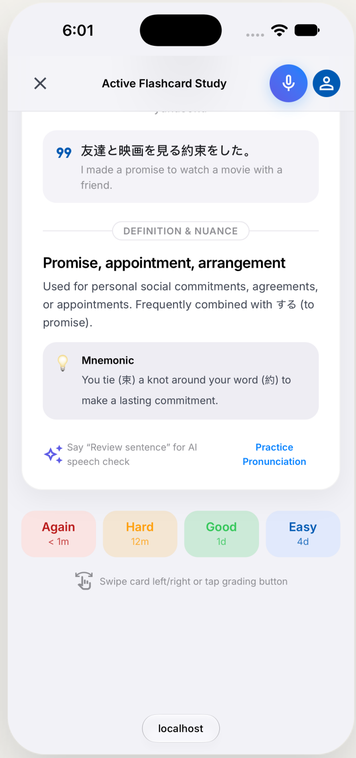
SHIPPED · QUICK WINS

The study screen, before and after

Before · card opens



Before · scrolled



Show Answer step before any grading

Grades pinned to the bottom, plus swipe

Due cards only, starting at the tapped card

One due count across every screen

"All caught up" end state

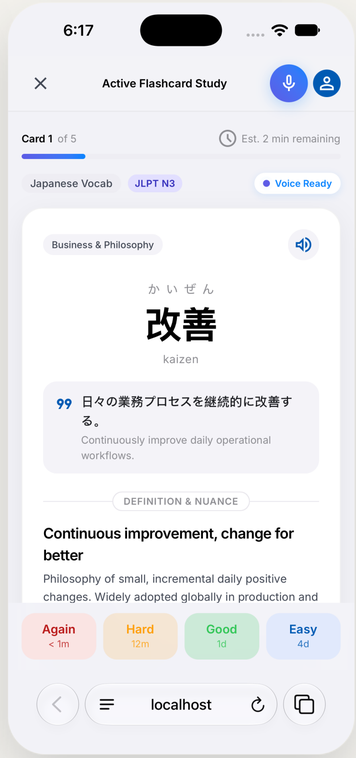
1 PR

8 files · +160 / −129 lines · verified on the iPhone simulator

After · card opens



After · revealed



PROPOSED V2 · TODAY

Open the app, see one thing to do

Even after the fixes, the home screen has 10 tap targets and none of them starts studying. Today replaces it with one answer and one button.

Due count and time estimate answer “how much today?”

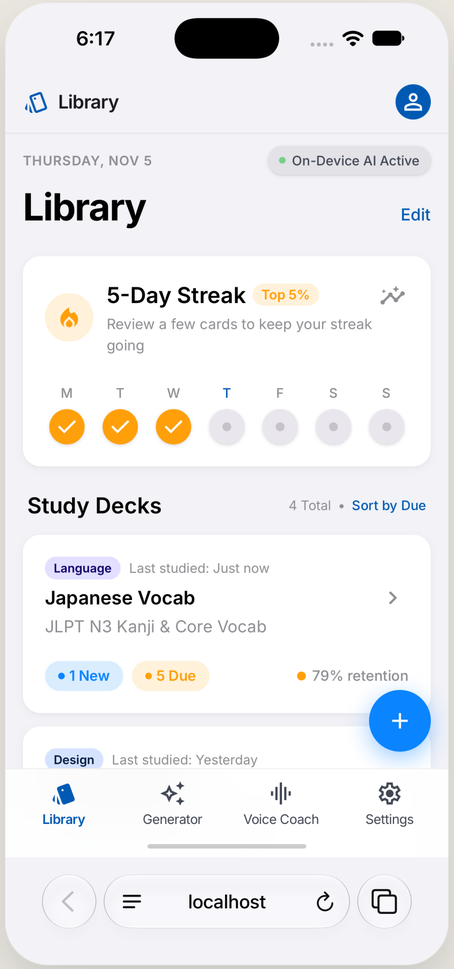
Start review runs every due card across decks

Decks become a list, one row each, still tappable

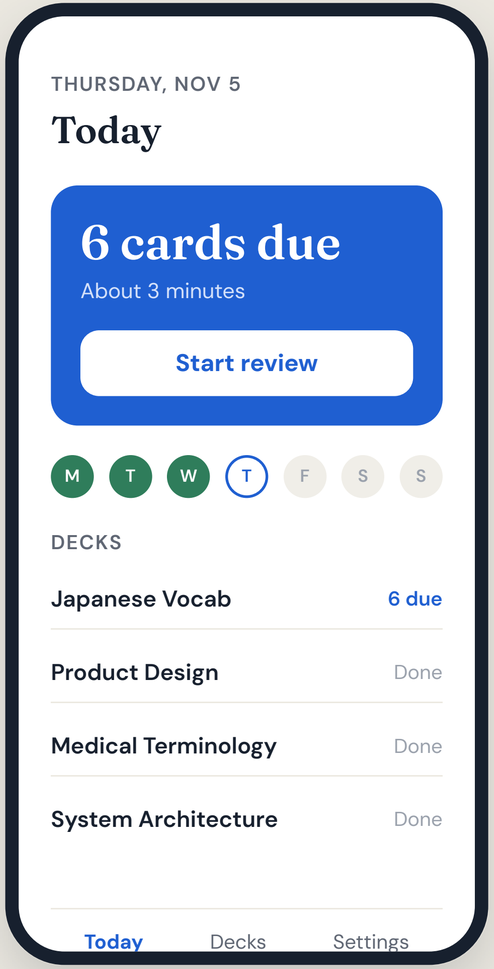
3 tabs, not 4: Today, Decks, Settings

Hypothesis: time from open to first card drops below 5 seconds.

Today, after quick wins



Proposed Today screen



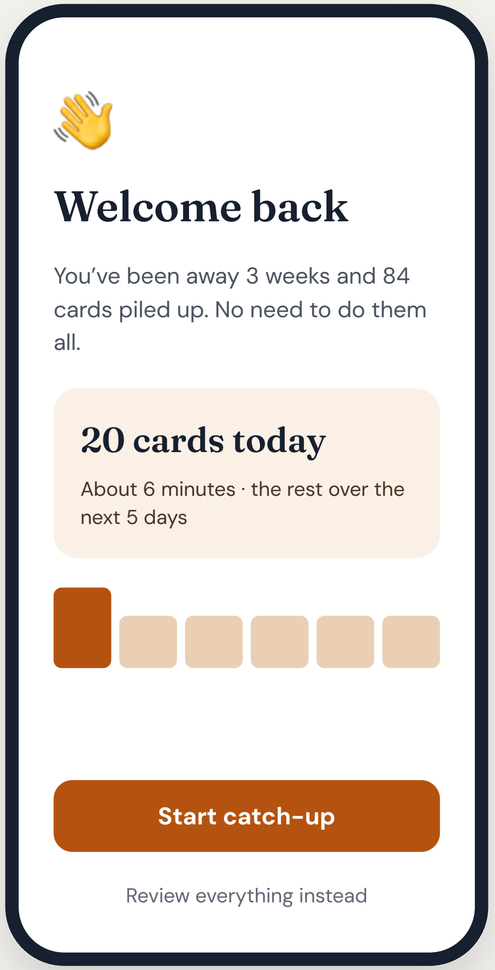
PROPOSED V2 · STRUCTURE

Fewer places to go, and a way back in

TODAY · 4 TABS	V2 · 3 TABS
Library	**Today + Decks**
Generator	Folded into **Add card › Generate**
Voice Coach	A **Speak answers** toggle in sessions
Settings	**Settings**

Hypothesis: capping the backlog at 20 raises return-session completion for learners away 7+ days.

Catch-up mode, for Priya



MEASURING SUCCESS

One north star, three inputs, one guardrail

NORTH STAR

Weekly active review days

Days per week a learner finishes at least one review. Spaced repetition only works if people come back.

INPUT · DANIEL

Time from open to first card

Target: under 5 seconds

INPUT · MAYA

Session completion rate

Sessions that reach “All caught up”

INPUT · PRIYA

Return completion after 7+ days away

And a second session within a week

GUARDRAIL

Voice and AI usage

Calmer UI must not bury the differentiators

Baseline: instrument first. Aura has no analytics yet, so there are no before numbers. Step one of v2 is logging these events.

WHAT'S NEXT

Validate the bets, watch the risks

Validate

- 5 interviews per persona to confirm or kill them
- Unmoderated test of the Today prototype
- A/B test the catch-up cap: 20 vs no cap

Risks

- Hiding voice and AI cuts discovery → offer a choice at first run
- A cross-deck queue blurs deck goals → keep per-deck study

Learnings

- Testing on a real device size found what the browser hid
- Trust breaks on numbers first
- Shipping quick wins earns room for bigger bets